ECON 0150 | Homework 3.4

Due: Friday, Oct. 30 at 5PM

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely. You are welcomed and encouraged to work in groups so long as your work is your own. Submit your figures and answers to Gradescope.

Q1. Study Hours

An economics professor claims that students spend 6 hours per week on average studying for their economics class. You survey 20 students and record the following number of hours they reported studying per week:

```
study_hours = [5.2, 7.1, 4.8, 6.5, 8.0, 5.5, 7.3, 4.2, 6.7, 5.8,
               7.5, 6.2, 5.0, 7.8, 6.3, 4.5, 6.9, 5.7, 7.2, 6.0]
```

Use an intercept-only regression (GLM) model to test the professor's claim. The p-value on the intercept tests whether the population mean is *zero*, so first subtract 6 from each student's hours. Then the intercept is the average difference from 6 hours, and its p-value tests the null hypothesis that this difference is zero. The Homework 3.4 Notebook includes this step:

```
study_hours_df['difference'] = study_hours_df['study_hours'] - 6
model = smf.ols('difference ~ 1', data=study_hours_df).fit()
```

a) What is the estimated difference in study hours from 6 hours?

b) What is the intercept's p-value?

c) In your own words, what does the p-value in this model represent?

Q2. Pennsylvania Household Income

According to USAFacts, the median household income in Pennsylvania was reported to be $73,824 in 2023. The Homework 3.4 Notebook loads a sample of median household incomes for 30 Pennsylvania counties. Use this sample to test whether the average median household income in these counties differs from the statewide figure.

a) Use an intercept-only model to test whether the average median household income in the sample differs from $73,824. As in Q1, subtract 73,824 from each county's median income before fitting the model. Report:

- Sample mean
- Standard error
- The p-value

b) Based on your results, does the sample provide evidence that the average county median income is different from the statewide figure? Briefly interpret your results in words.